

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Examination, 2017
CE 3107
(Elements of Solid Mechanics)

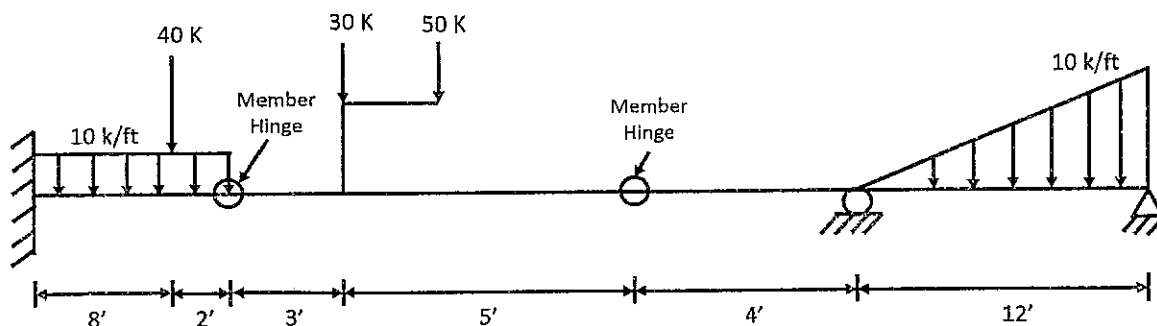
Full Marks: 210

Time: 3 hrs

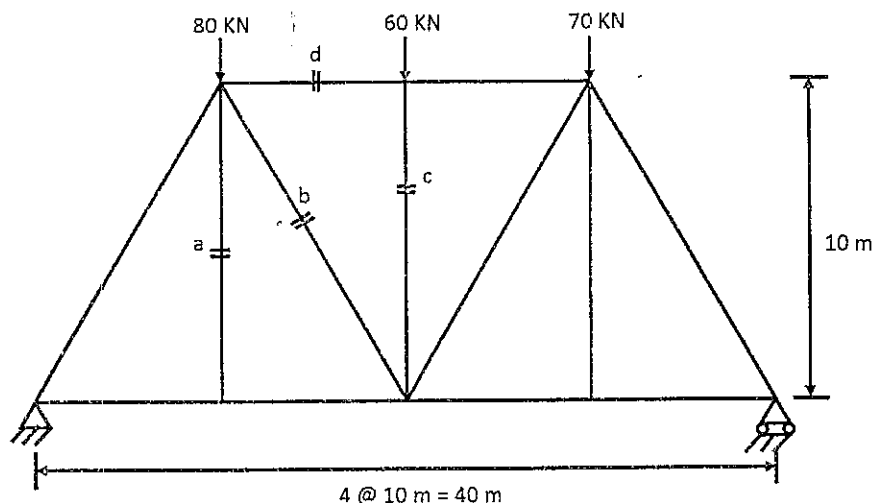
- N.B.** i) Figures in the right margin indicate full marks.
ii) Assume reasonable value for missing data.
iii) Answer any three questions from each section in separate script.

Section – A

- 1 (a) Define Shear force and Bending moment. Establish the relationships between load, shear and Moment. (10)
(b) What are the different types of Beams and Loadings? (05)
(c) Draw the shear force and Bending moment diagrams for the beam shown in figure below. (20)



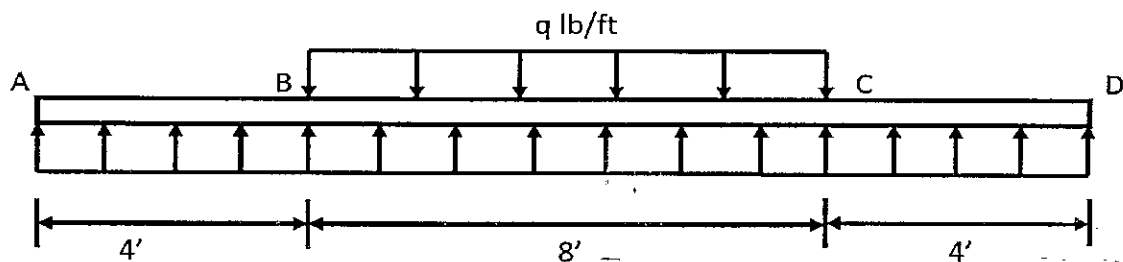
- 2 (a) Define the following terms: (06)
(i) Point of contra-flexure, (ii) Inflection Point, (iii) Dangerous section.
(b) The ultimate shear stress of a mild steel flat is 60 ksi. Calculate the diameter of the hole that can be punched through it without exceeding a compressive stress of 30 ksi, in the punch, thickness of the flat being $\frac{1}{2}$ inch. Also calculate the strain in the punch when, $E = 29 \times 10^6$ psi. (11)
(c) For the following truss calculate the stresses in members "a", "b", "c", and "d". The cross-sectional area of each member is 800 mm^2 . Also show whether the members are in Tension (T) or Compression (C). (18)



- 3 (a) What is meant by Flexural formula? Prove that, the flexural stress in any section varies directly with the distance of the section from the neutral axis. (12)
(b) Draw the stress-strain diagram of Mild steel and describe its different features. (08)

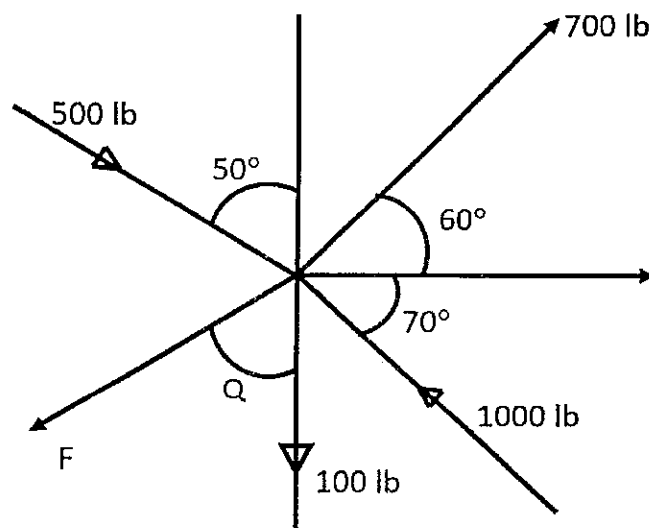
- (c) Following observations were made during the tensile test of a mild steel specimen: (15)
- Diameter of specimen = 30 mm
 Length of specimen = 200 mm
 Elongation at load of 60 kN = 350×10^{-4} mm (within proportional limit)
 Load at yield point = 290.6 kN
 Maximum Load = 395 kN
 Length of specimen after fracture = 230 mm
- Calculate the following terms:
- Modulus of Elasticity,
 - Yield Stress,
 - Ultimate stress, and
 - Percentage Elongation.

- 4 (a) Explain the term Poisson's Ratio. Define Hooke's Law. (05)
- (b) Obtain an expression for the elongation of a vertical bar of length, 'L', uniform cross-sectional area 'A', supported at its upper end and which is subjected to its own weight (W) only. (10)
- (c) A reinforced concrete foundation beam ABCD that supports a uniform load intensity q. (20)
 If the maximum shear force and bending Moment are 2000 lb and 2800 lb-ft respectively, then determine the maximum load intensity q. Assume factor of safety for shear and bending moment are 2.5 and 3 respectively.

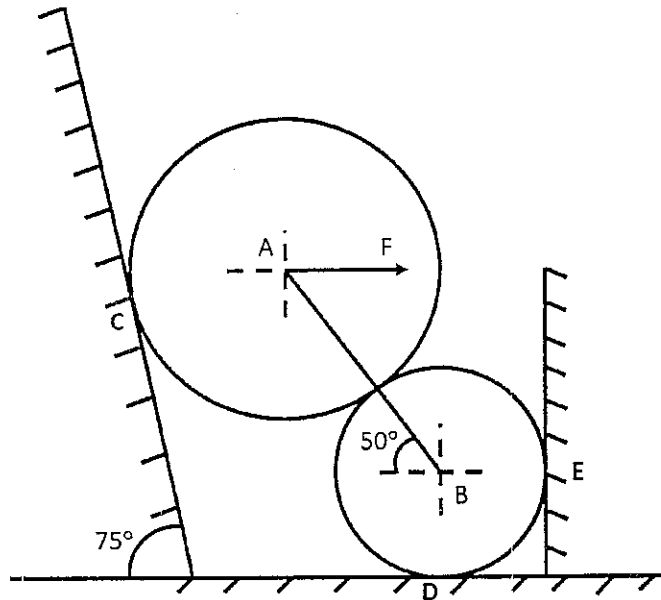


Section – B

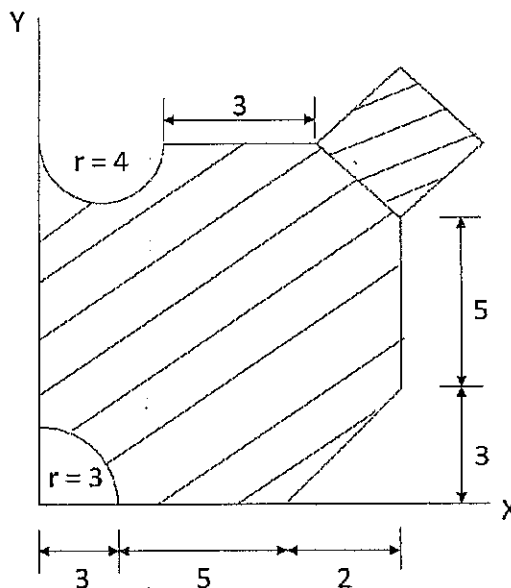
5. (a) Define: (10)
- Free body diagram,
 - Concurrent force system,
 - Two force member,
 - Equilibrium condition of a body.
 - Transmissibility of force system.
- (b) Find the value of F and Q of the following force system shown in figure below. Also find the resultant of the force system. (11)



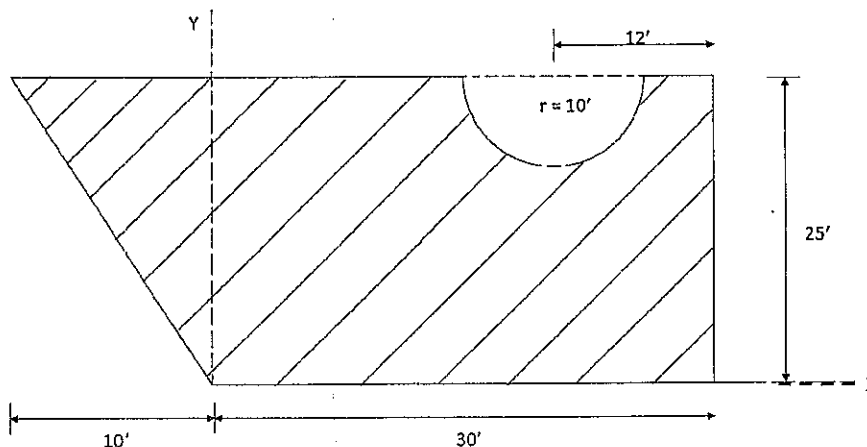
- (c) Two spheres are at rest against smooth surfaces, as shown below. Sphere A weighs 3000 lb and sphere B weighs 500 lb. Let $F = 1000$ lb and $Q = 75^\circ$. Find the reaction at C, D and E. (14)



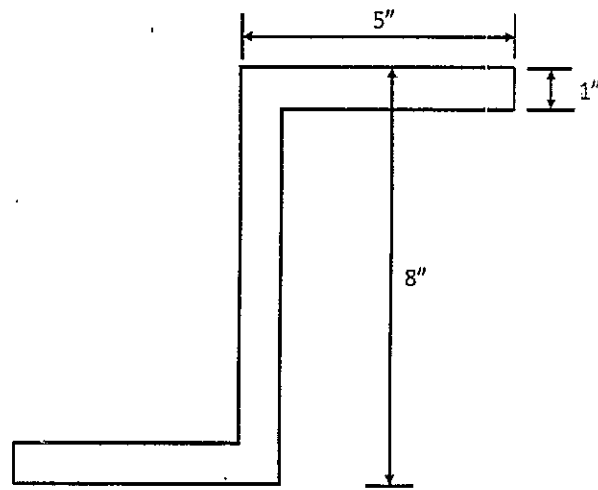
6. (a) What are the differences between centroid and center of gravity? Locate the centroid of an arc of a circle of radius r , which subtends an angle of 2β . (15)
- (b) Compute $\bar{x}$ and $\bar{y}$ of the shaded area shown in figure below. (20)



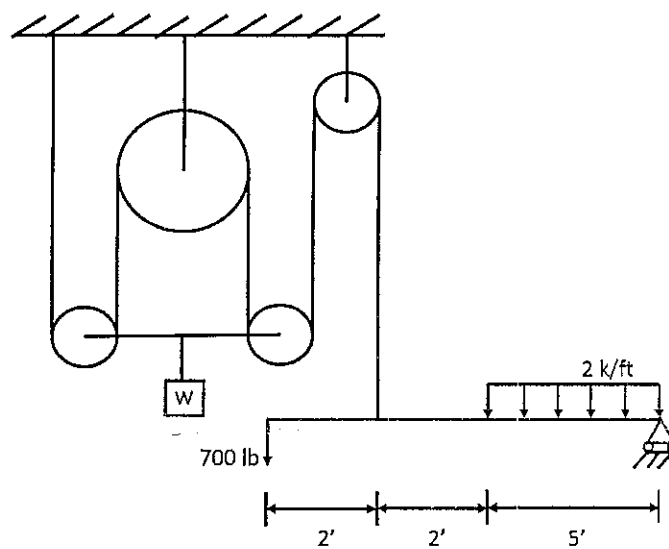
7. (a) Calculate the moment of inertia for the shaded area as shown in figure below about the Y-axis and the centroidal axis parallel to Y-axis. (18)



- (b) Prove that the product of inertia about centroidal axis is zero. Also find the maximum and minimum moment of inertia of Z-section of $8'' \times 5'' \times 1''$. (17)



8. (a) Write short notes on (i) Radius of gyration, (ii) Polar Moment of inertia. (06)
- (b) In the following figure, what will be the value of 'W' if equilibrium exists? Assume the pulleys are frictionless and weightless. (15)



- (c) Prove that the algebraic sum of moments of two concurrent forces about any axis perpendicular to the plane of the forces is equal to the moment of their resultant about the same axis. (14)

Khulna University of Engineering & Technology
Department of Urban & Regional Planning

BURP 3rd Year 1st Term Regular Examination, 2017.

CE 3151

(Fundamentals of Transportation Engineering)

Full Marks: 210

Time: 3 hrs

- N.B. i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Define Highway Geometrics. What are the elements of Highway Geometry? 05
(b) Draw neat sketch of a typical urban and rural road section with its major elements. 08
(c) Derive an expression for finding overtaking sight distance at road. 10
(d) The speeds of overtaking and overtaken vehicles are 100 kmph and 80 kmph, respectively on a two-way traffic road. If the acceleration of overtaking vehicle is 1.3 m/sec^2 and reaction time of 2 seconds. 12
Calculate – (i) Safe Overtaking Sight Distance.
(ii) Minimum Length of Overtaking Zone.
2. (a) What is meant by Stopping Sight Distance? Derive an expression for stopping sight distance on sloping ground. 12
(b) Define Camber. Is it necessary to provide camber on Horizontal Curve? Justify your answer. 05
(c) Why horizontal transition curves are provided at beginning and end of the sharp horizontal curves? Why excessive camber can not be provided on road pavement? 08
(d) A horizontal curve is to be designed for a section of a highway having a design speed of 80mi/h. Assume, $f_y = 0.14$. 10
(i) If the physical conditions restrict the radius of the curve to 600 ft., what value is required for the super elevation at this curve?
(ii) Is it a good design?
3. (a) What is meant by super elevation? Why is it provided? Derive an expression for super elevation on level road. 12
(b) Why extra width of road is provided on horizontal curve. 07
(c) Differentiate between Camber and Super Elevation. Why vertical curves are provided? Describe summit curve and valley curve. 10
(d) Define Gradient? What are the different types of gradient? 06
4. (a) Define following terms : 10
(i) Master plan
(ii) Shoulders
(iii) Driveways
(iv) Right of Way
(v) Curbs
(b) What is meant by PIEV? Why is it required? On what factors does it depend on? 10
(c) Describe the relationships between Land Use and Transportation. Write down different Land Use Models. 15

Section – B

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|----|-----|--|----|
| 5. | (a) | Define transportation engineering. Explain the benefits of good system of roads. | 10 |
| | (b) | What are the advantages and disadvantages of air transport? | 05 |
| | (c) | What is meant by road hierarchy? Classify roads according to their functions and define each of them. | 10 |
| | (d) | What is O-D survey? Write down the objectives of O-D survey. | 10 |
| 6. | (a) | Define Traffic Engineering. What are the causes of congestion? | 12 |
| | (b) | Write down the different physical measures of traffic engineering tools. | 10 |
| | (c) | Define the following terms – | 04 |
| | | (i) Bottleneck | |
| | | (ii) Flaring | |
| | (d) | How can conflicts be minimized? | 09 |
| 7. | (a) | What is meant by demand management? Write down the strategic measures of demand management. | 10 |
| | (b) | What is Capacity and Level of Service (LOS)? | 04 |
| | (c) | What are the roadway factors that affect capacity and LOS? | 06 |
| | (d) | What are the objectives of land-use planning and control? | 08 |
| | (e) | Draw a neat sketch of a cloverleaf inter-change. | 07 |
| 8. | (a) | Define the following terms: | 08 |
| | | (i) Parking Index | |
| | | (ii) Parking Volume | |
| | | (iii) Parking Load | |
| | | (iv) Park and Ride Scheme | |
| | (b) | Describe parallel and angle types of parking. Which type of parking is preferable for what type of situation or location? Explain. | 10 |
| | (c) | For angular parking determine the number of vehicles that can be parked for parking angle of 15° and length 80 m. | 07 |
| | (d) | What is meant by 'PCU'? What are the provisions for elderly and handicapped pedestrians? | 10 |

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2017

Hum 3117
(Accounting and Auditing for Planners)

Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is accounting? (05)
(b) State the users of accounting information. (10)
(c) Describe the steps in accounting process. (10)
(d) Explain the rules of debit and credit of accounts. (10)
2. Shasti opened her own consulting firm, Shasti Consulting on May 1, 2016. The following transactions occurred during the month of May.
May-1 Shasti invested TK 1,80,000 cash to start the firm.
May-2 Paid cash TK 18,000 for office rent for the month.
May-3 Purchased office supplies on account TK 15,000.
May-4 Paid TK 1,500 to advertise in the "Daily Bangladesh Protidin"
May-7 Received TK 45,000 cash for services provided.
May-10 Withdrew TK 10,000 cash for personal use.
May-12 Performed TK 40,000 of service on account.
May-15 Paid cash TK 30,000 for employee salaries.
May-18 Paid for supplies purchased on account on May 3.
May-20 Received TK 30,000 cash for services provided on account on May 12.
May-22 Borrowed TK 50,000 cash from the Bank on a note payable.
May-25 Purchased office furniture for TK 24,000 on account.
May-30 Utility bill paid in cash TK 2,000.
Instructions:
(a) Show the effect of the above transactions on the accounting equation/ tabular summery using the following column heading: Cash+ Account Receivable+ Supplies+ Office furniture= Account payable+ Note payable+ Shasti Capital. (22)
(b) Journalise above the transactions in the book of Shasti. (13)
3. Roshni is a licensed C. A. During the first month of operations of her firm, the following transactions occurred: (35)
2017
June-1 Roshni invested TK 200,000 cash.
June-2 Hired a secretary- receptionist at a salary of TK 10,000 per month.
June-3 Office rent paid in cash TK 9,000.
June-4 Stationary goods purchased in cash for office use TK 12,000.
June-15 Completed a tax assignment billed client TK 25,000 for services provided on account.
June-17 Received cash of TK 22,000 for services completed.
June-28 Paid secretary- receptionist TK 10,000 salary for the month.
June-30 Cash received from client for transaction June 15.
Required: Prepare necessary ledger accounts and balance show in to trial balance.
4. The trial balance of ABC Advertising Agency is shown below; the agency use an accounting period of one month. Thus monthly adjusting entries were recorded October 31, 2016:

ABC Advertising Agency Trial Balance October 31, 2016		
Account Titles	Debit (TK)	Credit (TK)
Cash	15,200	
Advertising Supplies	2,500	
Prepaid Insurance	600	
Office Equipment	5,000	
Notes Payable		5,000
Accounts Payable		1,200
Service Revenue		12,500
ABC, Capital		10,000
ABC, Drawing	500	
Salaries Expense	4,000	
Rent Expense	900	
Total TK	28,700	28,700

Additional information:

- i. Advertise supplies on hand at October 31 TK 1,000.
- ii. Expired insurance for the month is TK 100.
- iii. Salaries unpaid at October 31, TK 1,200.
- iv. Depreciation on equipment for the month is TK 50.

Instructions:

- (a) Prepare a statement of comprehensive income for the month ending October 31, 2016. (15)
- (b) Prepare an owner's equity statement. (05)
- (c) Prepare a statement of Financial Position as on October 31, 2016. (15)

Section – B

5. (a) What is tax? (05)
- (b) What are the purposes of taxation? (10)
- (c) State the cannons of taxation. (10)
- (d) Distinguish direct tax and indirect tax. (10)
6. (a) Define internal audit. (05)
- (b) State the features of internal audit. (10)
- (c) What are the objectives of internal audit? (10)
- (d) Discuss the advantage of internal audit. (10)
7. (a) What do you mean by internal control? (05)
- (b) State the scope of internal control. (10)
- (c) What are the objectives of internal control? (10)
- (d) Discuss the limitations of internal control. (10)
8. Write short notes (any five): (5×7) (35)
 - a) Audit Programme
 - b) Interim Audit
 - c) Audit Risk
 - d) Tax-holiday under the income tax Act
 - e) Voucher
 - f) Audit Evidence

Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Regular Examination, 2017.
URP 3161
(Housing and Real Estate Development)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Figures in the right margin indicate full marks.
iii) Answer any **three** questions from each section in separate script.

Section – A

1. a. What do you understand by "Housing Rights"? How are housing rights guaranteed by international organization and why do we need them? (10)
b. What are the major components of a functional housing market? Explain all of those components with example. (12)
c. "Adequate housing is determined in part by social economic, cultural, climatic, ecological and other factors"- explain with examples. (13)
2. a. Is housing a need or demand? How is this need defined in an environmental system? (11)
b. About 85% of the total housing in Bangladesh is from informal sector. Why? What are the impacts of such informal system on the market and government? (12)
c. "Urban areas are segregated geographical space". How do the poor find their place in such a segregated space? Explain. (12)
3. a. What are the major sources of sectoral housing in Bangladesh? Which sectoral housing is the most important according to the National Housing Policy (present) of Bangladesh and why? (12)
c. Explain the World Bank (WB) influenced decadal changes in the housing policies in Bangladesh. What is the present approach of the Bangladesh Government to the massive housing demand? (10)
d. What was the WB condition for site and services? How and why Government site-and-services failed and which component was most significant for the failure? (13)
4. a. Explain the hedonic approach for determining housing price? How can non-market factors affect housing prices of Bangladesh? (14)
b. What are the major housing challenges for urban Bangladesh? Explain the existing policy and regulatory framework for the urban sector. (10)
c. Explain the urban housing strategies of the 7th five year plan FY 2016-2020. (11)

Section – B

5. a. What are the fundamental ways to use the word real estate in empirical and professional field? How is real estate market activity influenced by the activities and conditions that take place in three sectors of a market economy? (12)
b. Identify three ways in which real estate markets differ from the market for publicly traded stocks? (15)
c. State the strategies of managing real estate investment risk. Hospitality Property is the most uncertain in terms of cash flow-Why? (08)

6. a. Explain and Give the specific examples of "Improvement on the land" and "Improvement to the land" in relation to real estate development. (07)
 - b. Write down the desired supply side characteristics of Urban Growth. Explain the concept of Industry Economies of Scale and Agglomeration Economies. (10)
 - c. How does each of these affect the slope of a bid-rent curve: Faster travel, more frequent trips, more commuters, higher-density housing, and high hourly wage rate? (08)
 - d. Explain the monetary and non-monetary benefits and costs for a real estate investment with examples. (10)
7. a. What is market research? Write down the importance of market segmentation in real estate market research. (10)
 - b. Define consumer surplus and market surplus. Quantities demanded and supplied are expressed as $Q_d = 80 - 5P$ and $Q_s = -24 + 8P$ respectively. Find out the market surplus from these equations with necessary diagrams. (12)
 - c. Exemplify the interaction of real estate value determining sectors with a diagrammatic representation. (13)
8. a. Explain the market trends that affect a real estate market with necessary examples. (12)
 - b. An investment opportunity having a market price of Tk. 1,000,000 is available. You could obtain a Tk. 750,000, 25-year mortgage loan requiring equal monthly payments, with interest at 7.0 percent. The following operating results are expected during the first year: (10)

Effective Gross Income	Tk. 200,000
<u>Less Operating Expenses and CAPX</u>	<u>Tk. 100,000</u>
Net Operating Income	Tk. 100,000

- i. For the first year only; determine the:
 - ii. Gross Income Multiplier.
 - iii. Operating expense ratio,
 - iv. Monthly and annual mortgage payment,
 - v. Debt coverage ratio,
 - vi. Overall cap rate,
 - vii. Equity dividend rate.
- c. The following information of a real estate property is given below: (13)

Units	20
Market rent per unit	1500
Other income (Affected by vacancy)	20,000/year
Other income (Not affected by vacancy)	10,000/year
Vacancy rate	12% of GPI
Operating expenses	50% of EGI
Purchase price	1,500,000
Market cap rate	7%

Find out the followings:

- i. What is the Effective Gross Income (EGI)?
- ii. What is the NOI?
- iii. What is the Cap Rate?
- iv. What is the Market Valuation?
- v. What Equity do you have upon buying this property?

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 1st Term Final Examination, 2017
URP 3121
(Rural Development Planning-I)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define rural development. What are the objectives of rural development? (12)
- (b) As a Planner of Rural Development Academy (RDA) elaborate the importance of rural development in Bangladesh. (13)
- (c) Why decentralization and participation is essential for rural development? (10)

2. (a) Define GDP and GNP. Discuss Expenditure Approach and Income Approach for calculating GDP. (13)
- (b) What is Physical Quality of Life Index (PQLI)? Calculate PQLI for the following countries and explain the result. (12)

Country	Life expectancy at age one	Infant mortality	Literacy
	Year	Per 1000 live birth	Percent
P	55	175	27
Q	81	51	83
R	67	79	31

- (c) Write a critique of Human Development Index (HDI) as a measure of development. (10)
3. (a) "Poverty measure is a function of individual income and poverty line" – explain the statement (07)
 - (b) What are the advantages and disadvantages of Squared Poverty Gap (SPG) Index? (05)
 - (c) Determine (i) Headcount Index (ii) Poverty Gap (PG) Index and (iii) Squared Poverty Gap (SPG) Index for the following countries and explain the results. (13)

Expenditures for each individual in countries				
Country P	3700	4900	6500	8100
Country Q	2500	4500	4700	5700
Country R	2600	4600	5000	6500

Assume the poverty line is 5000.

- (d) What are the policy implications of different poverty measures? (10)
4. (a) "Agriculture has a multitude of functions in the development process"- Explain them with appropriate examples. (13)
 - (b) Write down the Strength and Threat of the agriculture sector under the analyzed SWOT in the National Agriculture Policy, 2013. (12)
 - (c) Mention the opportunities and constraints considered for framing out the National Agriculture policy 1999. (10)

Section – B

5. (a) Write about the basic four principles of defining a rural area. How will you demarcate a rural area following the principles? (10)
- (b) "Different factors influence rural settlement patterns in different ways" – Clarify the statement with necessary examples and sketches. (10)
- (c) Make a comparative analysis between the benefits of two major patterns of rural settlement with necessary examples. (10)
- (d) Mention the factors that governed farm sizes. (05)
6. (a) Define the concept of rural centre. (05)
- (b) What are the Procedures for Drawing up a Hierarchical Rural Centre Plan? "Highest level growth foci require higher threshold population". - Explain the statement with appropriate examples. (10)
- (c) Which is the best method for determining the present centrality of settlements for small area? – Use the example of health service in this regard. (10)
- (d) How do you identify suitable industries in a rural area by using industrial development screening matrix? (10)
7. (a) What are the main components of "Comilla Experiment"? Make a critical analysis on the contribution of "Comilla Experiment" in rural development of our country. (12)
- (b) Describe the activities that need to be changed to improve the quality of life of rural people especially those who are of various categories. (07)
- (c) Write down rural development ideas and approaches during the period 1950 to 2000. (10)
- (d) Give a short description on the success of V-AID programme of rural development. (06)
8. (a) Write short notes from the following (any five): (5*7= 35)
- (a) FFWP
 - (b) Governmental Organizations involved in Development Activities
 - (c) Local skill based industries
 - (d) Spatial planning for Rural Areas
 - (e) Housing patterns in Bangladesh
 - (f) RWP
 - (g) Types of rural industries
-